CloudCrop

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Introduction

CloudCorp takes the guesswork out of plant care by combining environmental sensors data with weather forecasts to automate watering using AI decision-making. The system continuously monitors soil moisture, temperature, and sunlight levels around the garden. The collected data is sent to the Raspberry Pi for processing. The AI decision-maker uses the processed data along with weather forecast data to make smart watering decisions, including skipping watering when there is rain forecast within the next few hours, saving water and preventing over-saturation. All of this data is logged and displayed on a web dashboard that the users can access from their phone or computer, giving the users complete insight of their plants growing conditions anytime from anywhere. Whether you are a busy professional, frequent traveler, an elderly person, new to gardening or simply want to spend less time worrying about your plant, this system keeps your plants healthy and thriving with minimal effort.

Hardware

Soil Moisture Sensor SEN0114: Determines the water content in the soil. The sensor sends the data to the Raspberry Pi. The Raspberry Pi determines whether the plants need watering or not based on the moisture level.

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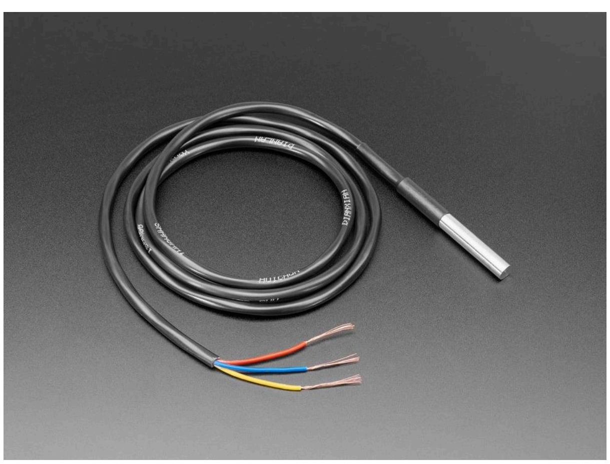
I2C IP68 Waterproof Ambient Light Sensor: Measures the sunlight exposure to the plants and forward the data to the Raspberry Pi. The data received from the sensor helps the system to manage watering. The waterproof sensor helps us get correct readings at all time.

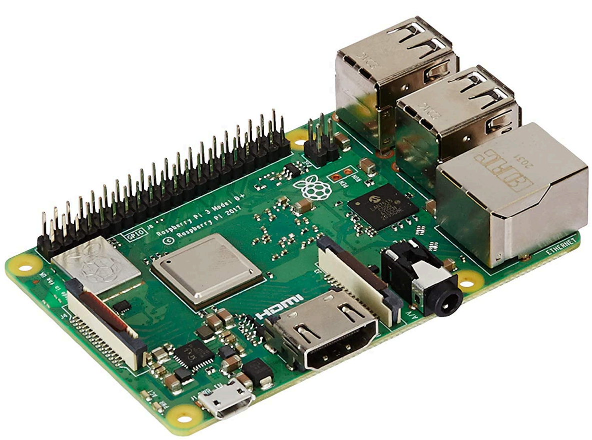
Waterproof DS18B20 Digital Temperature Sensor: Monitors the temperature around the plant. Temperature data helps the system make smarter watering decisions, since plants need more water in hot conditions and less in cool conditions. The waterproof casing ensures accurate readings in extreme conditions.



Water Pump 5V: Delivers water from the water container to the plants when the system determines watering is needed.

Vinyl Tubing: Connects the water pump to the garden beds or individual plants, directing water exactly where it's needed. The flexible tubing can be routed around your garden and positioned near plant roots for efficient, targeted watering.

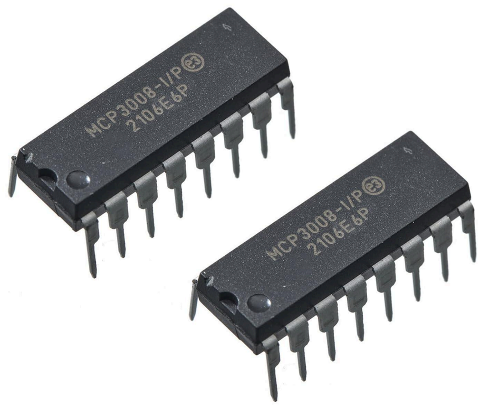
Raspberry Pi 3B+ is the brain of the system that reads all the sensor data, checks weather forecasts, makes watering decisions, and controls the pump. The data is also sent to the web dashboard so the users can monitor their plant's condition from anywhere.



5V 3A Power Cable: Provides reliable power to the Raspberry Pi and water pump. The 3A (amp) rating ensures the Raspberry Pi has enough power to run continuously while operating the pump when needed.

Male to Female Converter: Connects the sensors and pump to the Raspberry Pi. These adapters make it easy to wire everything together without soldering, allowing for simple setup and the ability to disconnect components if needed for maintenance or troubleshooting.

MCP3008 Analog-to-Digital Converter: Allows the Raspberry Pi to interact with the soil moisture, light, and temperature sensors by converting the analog voltage signals into digital values. Without this product, the Raspberry Pi would be unable to read any values from the sensors.

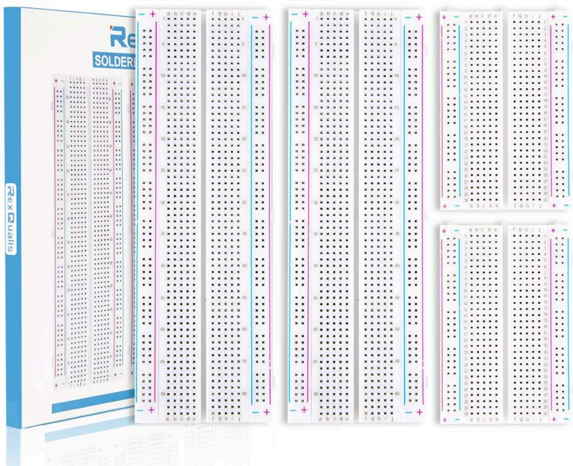


4 Channel Optocoupler Relay Module: Acts as a safe switch that allows the Raspberry Pi to control higher-powered devices like the water pump. The optocoupler protects the Pi from electrical damage by isolating it from the pump's power circuit, while the 4 channels allow control of multiple pumps or valves for different garden zones.

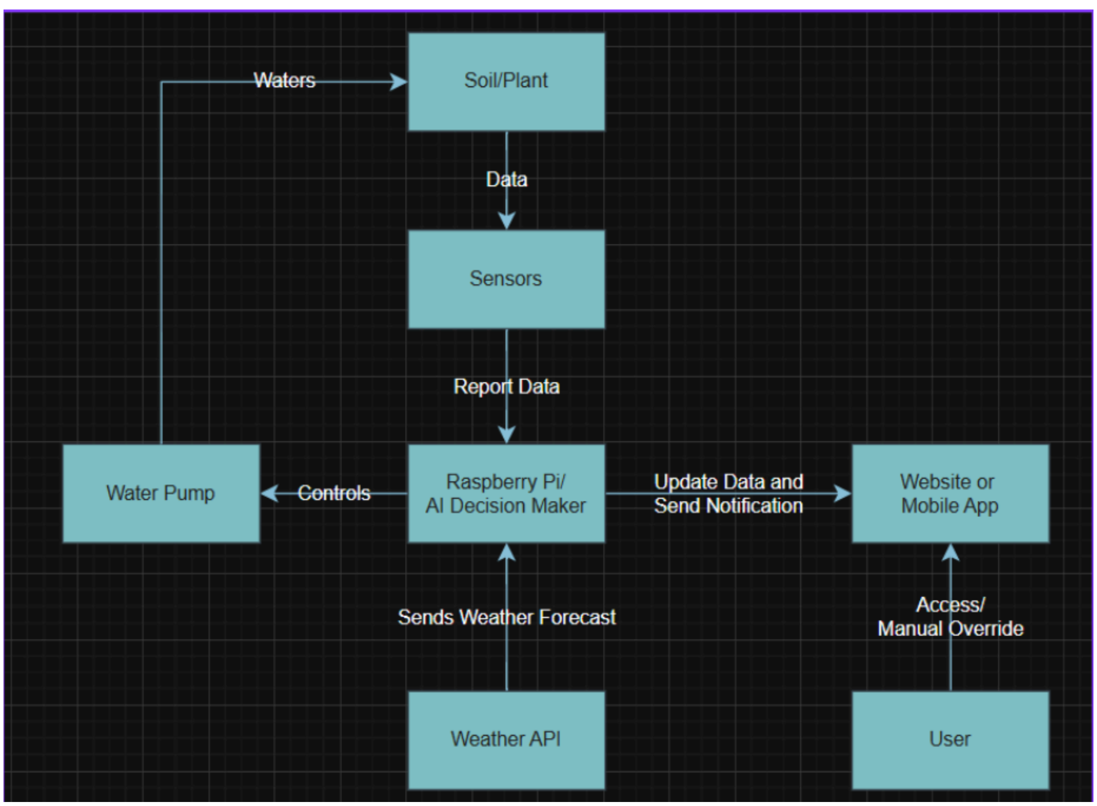
12V 5A Power Supply: Powers the water pump with the higher voltage and current it needs to operate effectively.

Jumper Cables (Male-to-Female, Male-to-Male, and Female-to-Female): Connect all the components together, linking sensors to the Raspberry Pi. The different connector types allow to wire between various components regardless of their pin configurations, making assembly flexible and straightforward without any soldering required.

Breadboard: Provides a reusable platform for organizing and connecting multiple wires and components in one central location. It makes the wiring neater and easier to troubleshoot, and allows modification without permanent changes.



Software Architecture



The proposed system uses a smart irrigation architecture design to automatically water plants based on real-time sensor readings. The system combines multiple sensors, artificial intelligence (AI) based decision-making, and web application to monitor and control watering. The core of the system is the Raspberry Pi 3B+ and the AI Decision Maker, which acts as the central controller. Raspberry Pi collects the soil data from the sensors, analyzes this information using AI models, and controls the water pump automatically. The goal of the system is to maintain proper sun exposure and moisture levels to support healthy plant growth without wasting water.

Data Collection Layer

A group of sensors is placed in the soil and around the plants to continuously monitor:

- **Soil Moisture** – to determine the amount of water in the soil.
- **Temperature** – to measure the surrounding temperature of the plant.
- **Sunlight** – to monitor sunlight exposure level to the plant.

These sensors collect and report data to the Raspberry Pi in real time. The AI uses this gathered data to make its decisions.

A.I. Decision-Making and Control Layer

The Raspberry Pi / AI Decision Maker serves as the processing hub of the system. It:

- Collects the data from the environmental sensors.
- Retrieves additional weather information from an external Weather API.
- Analyzes the data to determine whether watering is needed.
- Sends control signals to the Water Pump to start or stop the water flow.

We are planning on testing two A.I. algorithms. The first will be a Random Forest and the second will be an XGBoost Model. The model will need to have regression so we can let the pumps know how long to water the plant for. We will be using ChatGPT to generate mock data for us.

- Random Forest

    - Uses multiple decision trees.
    - The output is given as an average of all the decision trees.
- XGBoost (eXtreme Gradient Boosting)
    - Also uses multiple decision trees but each tree corrects the errors of the previous tree.
    - Reiterates until error meets a certain minimum threshold.

This intelligent control ensures water is delivered only when necessary, preventing overwatering the plants and reducing water waste.

Communication and User Interface Layer

The system continuously updates and pushes the processed data to the web application and stores it in the database. The web application offers users an intuitive way to monitor and control the entire system remotely, providing complete visibility and operational flexibility from anywhere in the world with internet connection. Users can:

- View real-time data such as soil moisture, temperature, and sunlight levels.
- Receive notifications and alerts regarding abnormal sensor readings and watering activity.
- Access previous data and performance reports for analysis and decision-making.
- Manually override the system to perform manual operations when necessary.

The front-end interface will be developed using HTML for structure, CSS for styling and responsive design, and JavaScript for dynamic content and interactivity. The back-end services, responsible for data storage, decision logic, and communication with the hardware, will be developed primarily in Python.

The proposed software architecture runs on a continuous cycle that collects data, analyzes it, and makes smart decisions. By combining environmental sensors, AI-driven decision-making, cloud-based weather data, and an intuitive user dashboard, the system delivers a robust, efficient, and user-friendly solution to keep the plant healthy.

Expected Project Results

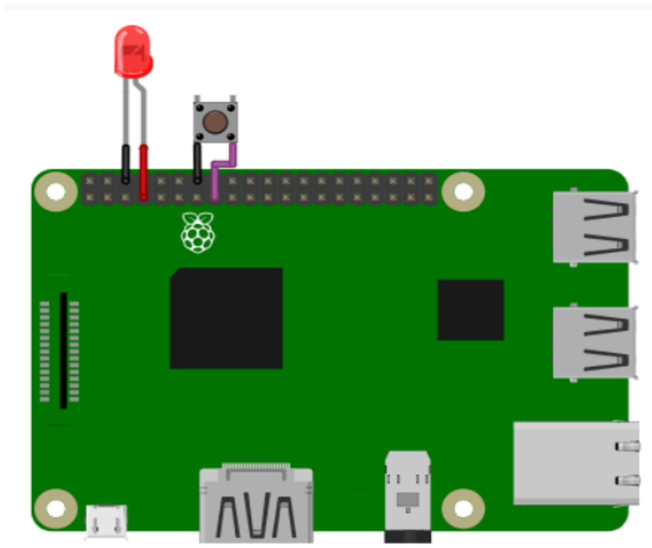
The expected deliverable is a product that implements a collection of 3-4 sensors that monitor soil moisture, temperature, and light conditions to reduce the need for manual plant maintenance. The product will also have control over 2-3 water pumps that are either automatically (by an A.I. decision maker) or manually activated (using the integrated app or website) through a relay module. The user will be able to select which species of plant they want the system to water so as to assure no overwatering. In addition to the sensors, the system will also be connected to a weather api where it will be able to see the upcoming forecast. If rain is predicted in the next 5 hours, it will hold off on watering the plants. Overall, the project aims to demonstrate how affordable hardware and intelligent software integration can significantly reduce water waste, improve plant health, and automate greenhouse maintenance for small-scale growers, researchers, and hobbyists.

The measures of success for this project would be accurate soil monitoring within a 5% accuracy when compared to a calibrated reference meter, watering pumps only activating when soil moisture goes below a certain threshold, accurately skip watering when rain is predicted at most 6 hours in advance with 80% accuracy, and save 20-30% water over a two-week period when compared to manual watering.

Simulation

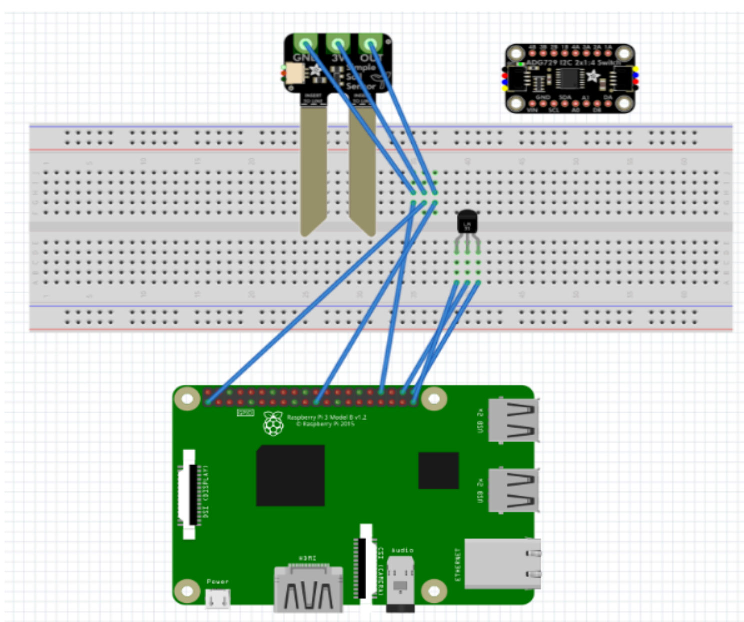
Wyliodrin Simulation: This is a tool used for simulating Raspberry Pi and we have been using it to begin understanding how we can code on our Raspberry Pi once we get it. Unfortunately, this program only allows coding in Javascript so we are looking for a different simulator.

```
// Load the library onoff.Gpio
var Gpio = require("onoff").Gpio;

// Here we define the variables which control the
// pins on the RaspberryPi.
var led = new Gpio(4, "out");
var button = new Gpio(22, "in");


// We stay in this while loop until the button is
// pressed. As you can see, we read the value of
// the pin associated to the button every 1 second.
while(button.readSync() === 0) {
    sleep(2000);
    led.writeSync(1);
    sleep(2000);
    led.writeSync(0);
}

console.log("onoff.Gpio tutorial finished!");
```



Fritzing

Fritzing is the tool that we are using to create the Raspberry Pi schematics that we can then upload to Wyliodrin. As you can see here, I was able to find a soil moisture sensor and a light sensor.



Budget for Prototype

Component	Unit Price	Quantity for project	Total Price
Soil moisture sensor	$2.70	4	$10.80
Gravity I2C waterproof light sensor	$8.90	4	$35.60
Waterproof Temperature sensor	$9.95	4	$39.80
Water pump 5V	$9.99	3	$29.97
Vinyl Tubing	$5.59	3	$16.77
Raspberry Pi 3B+	$47.00	1	$47.00
Breadboard	$6.63	1 pack	$6.63
MCP3008 Analog-to-Digital Converter	$13.99	1	$13.99
4 Channel Optocoupler Relay Module	$6.88	1	$6.88
Female to Male converter	$4.65	1	$4.65
Jumper Cables	$5.58	1 pack	$5.58
MicroSD Card	$7.99	1	$7.99
MicroSD to SD Adapter	$4.99	1	$4.99
12V 5A Power Supply	$8.99	1	$8.99
Raspberry Pi Power Cable	$8.99	1	$8.99
		Total Price =	$248.63

Business Plan

We assumed a 6 month development and 6 month selling business plan for this project. With the assumption that we would be making and selling 10,000 products, here is a breakdown of all the associated costs including; Hardware, Manufacturing and Assembly, Labor, Overhead, Patents/Licenses, Marketing, and Maintenance/Warranty.

Hardware	**Quantity**	**Cost per unit**	**Cost per 10,000 units**
Soil moisture sensor	4	$10.80	$108,000.00
Gravity waterproof light sensor	4	$35.60	$356,000.00
Waterproof Temperature sensor	4	$39.80	$398,000.00
Water pump	3	$29.97	$299,700.00
Vinyl Tubing	3	$16.77	$167,700.00
Raspberry Pi 3B+	1	$47.00	$470,000.00
Breadboard	1	$3.00	$30,000.00
MCP3008 A/D Converter	1	$7.00	$70,000.00
Relay Module	1	$6.88	$68,800.00
Female to Male converter	1	$4.65	$46,500.00
Jumper Cables	20	$1.00	$10,000.00
MicroSD Card	1	$4.00	$40,000.00
Power Supply	2	$6.50	$65,000.00
	Hardware Total=	**$212.97**	**$2,129,700.00**

Manufacturing/Assembly	**Cost per 10,000 units**	**Notes**
PCB fabrication and assembly labor	$80,000	small-run line at ~$8/unit
Product assembly (housing, wiring, testing)	$120,000	~$12/unit average labor
Quality assurance testing & packaging	$50,000	~5% of manufacturing cost
Shipping from factory to warehouse	$40,000	bulk container shipping
Subtotal (Manufacturing)	**$290,000**	

Labor	Cost per 10,000 units	Description
Hardware design & prototyping	$60,000	1 engineer for 6 months
Software & firmware development	$120,000	2 devs for 6–8 months
Cloud dashboard / app development	$100,000.00	1 front and back-end dev
Testing, calibration, and validation	$30,000.00	QA engineers + field testing
Project management, documentation	$40,000.00	Coordination, specs, reports
Subtotal (R&D labor)	**$350,000.00**	**One-time development**

Overhead	Cost per 10,000 units	Notes
Office/workspace rent, utilities	$60,000	1 year small facility
Insurance (liability, property)	$30,000	Required for electronics sales
Legal & accounting services	$25,000	Business setup, contracts, filings
Administrative salaries	$70,000	CEO, operations, admin staff
Subtotal (Overhead)	**$185,000**	

Patents/Certifications/Licensing	Cost per 10,000 units	Notes
Utility Patent (Smart irrigation logic)	$15,000–$25,000	U.S. filing and legal fees
Design Patent (optional)	$5,000	Protects visual design
FCC / CE Certification	$10,000–$15,000	Required for electronics
Subtotal (IP & Compliance)	**$35,000–$45,000**	

Marketing, Sales, and Distribution	Cost per 10,000 units	Notes
Branding, packaging design	$20,000	Logo, box art, documentation
Website, e-commerce setup	$15,000	Storefront + hosting
Digital marketing campaigns	$80,000	Ads, influencers, SEO
Retail partnerships & logistics	$40,000	Distribution contracts
Trade shows & product demos	$15,000	Booths and travel
Subtotal (Marketing & Sales)	**$170,000**	

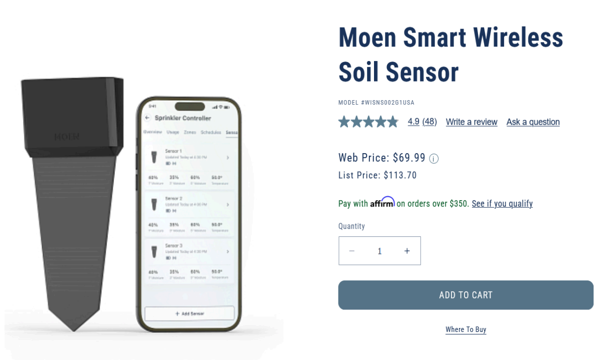
Maintenance and Warranty	Cost per 10,000 units	Notes
Customer support team	$40,000	Email/chat for 1 year
Warranty replacements (2% failure rate)	$189,750	200 units × $167.09. avg cost + labor
Repair/refurbishing logistics	$25,000	Shipping + labor
Subtotal (Maintenance)	**$254,750**	

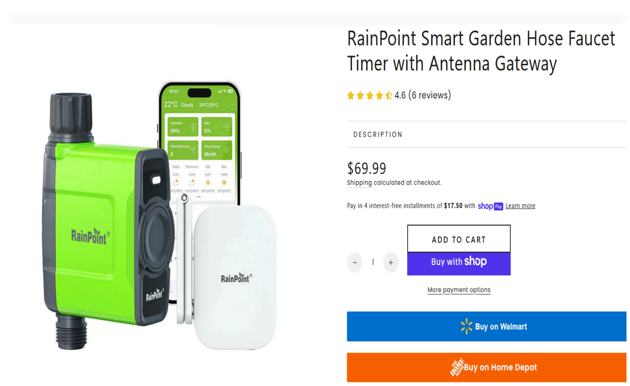
Final Estimate	Cost (USD)
Hardware	$2,129,700
Manufacturing & Assembly	$290,000
Labor	$350,000
Overhead	$185,000
Patents	$40,000
Marketing, Sales, and Distribution	$170,000
Maintenance & Warranty	$254,750
Total Estimated Cost	**$3,419,450**

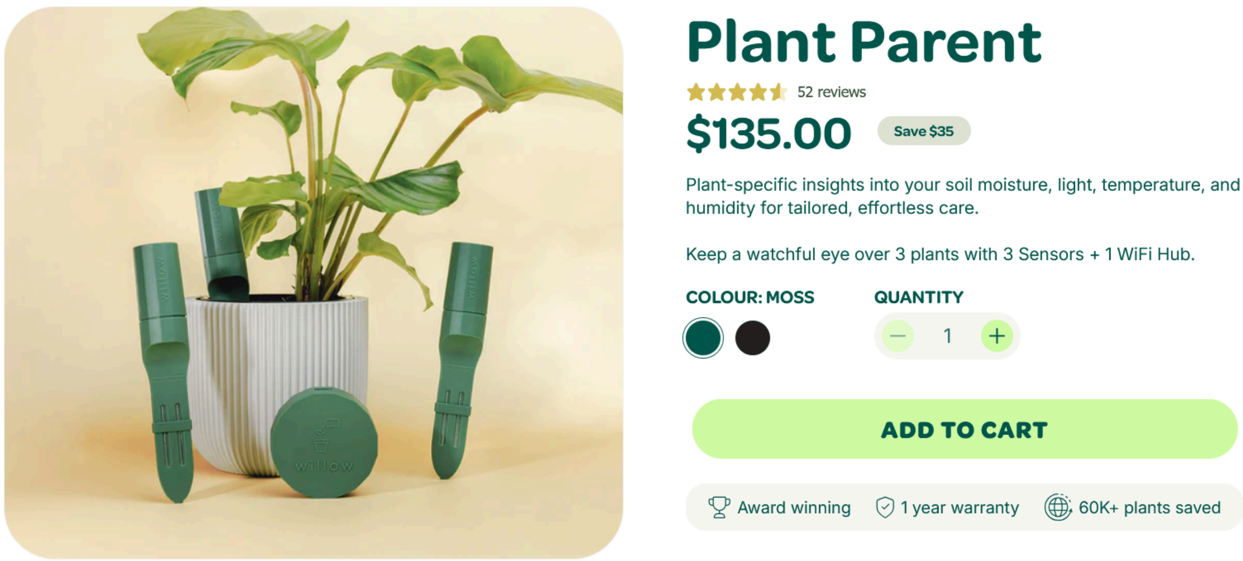
Our total estimated cost comes out to $3,419,450 meaning we would have to sell 10,000 products at $342 each to break even. These estimated costs are very liberal and it is likely that the costs would be much lower in reality. One example is that the hardware costs are all done at stock prices. If we were to buy 10,000 of each, there would be bulk order deals and the overall cost would go down.

Competitors

Product	Core Functionality	Key Features	Limitations	Price
CloudCrop 3 Sensors	AI-driven smart gardening and automated watering system	AI synthesizes soil moisture, sunlight, temperature, and weather forecasts. Delivers all-in-one plant care with automated watering.	Higher initial price due to advanced AI integration and built in water pump for autonomous watering.	$360.00
Moen Smart Wireless Soil Sensor Moen Link	Soil moisture monitoring for irrigation optimization	Measures soil moisture; can work with Sprinkler Controller to adjust watering schedules	Requires Sprinkler controller for automation; No light or humidity sensing; Lack AI integration.	$69.99
Willow 3 Sensors & 1 Hub Willow Link	Indoor environmental monitoring.	Tracks soil moisture, temperature, light, and humidity through a mobile app	Does not automate watering; focuses on monitoring and alerts; indoor use only.	$135.00
RainPoint Garden Hose Timer RainPoint Link	Programmable hose-based watering system	Two independent valves with programmable schedules; watering-skip based on weather forecasts	No sensor; Limited to hose-based outdoor watering with timer	$69.99







Our competitor research shows that CloudCrop has a strong market position through its combination of AI integration and unique features. While competitors like Moen offer wireless soil sensors to collect environmental data, Rainpoint has a hose-based watering system with timer, and Willow focuses on indoor plant monitoring with their temperature and soil sensors, no competitor provides an autonomous plant care. All competitors mainly rely on pre-set thresholds and simply notify users to manually water their plants, which can lead to over or under watering the plant. CloudCrop provides a complete solution to all the problems. It combines all the sensor data and weather forecast data with AI-driven decision-maker to automatically water the plants only when necessary. CloudCorp continuously learns and optimizes plant care, rather than just monitoring conditions or following fixed schedules. This makes CloudCorp a truly intelligent and resource-efficient product for gardening.

Targeted Audience

The target audience for CloudCorp primarily includes busy professionals, vacation or traveling homeowners, older gardeners, small-scale farmers, and aspiring gardeners. Busy professionals and frequent travelers who do not have enough time, are always away from home and struggle to water their plants can use this system to automatically water and monitor their plants . Older gardeners who do not have physical strength to take care of their plants, can benefit from this autonomous system. Aspiring gardeners and beginners who are not sure how much water or sunlight a plant needs, can take advantage of the system to learn about the right amount of water and sunlight for their plants. Finally, small-scale farmers can use the system to reduce water usage and save on maintenance costs.

Marketing

CloudCrop's marketing strategy will use different approaches to reach different groups of users. Our main strategy is to use our product website, which will be regularly updated and offer discounts and promotions. Secondly, we will utilize digital platforms like Facebook, Instagram, TikTok, and YouTube to promote our product using ads and short videos. For direct marketing, we will attend trade shows, conferences, and local farmers markets to showcase our product and engage directly with gardeners and homeowners. We will collaborate with influencers to use our product and give their reviews on it. By using a multi-approach strategy, we hope to connect with every potential customer.

Timeline

Week	Tasks
Sep	- Research, features, system design, and assign team roles.
Oct	- Propose a project idea and presentation on it.
Oct	- Research hardware and finalize hardware list.
Nov	- Learn about AI models and pick the right one for our product.
Nov	- Confirm hardware order and estimated delivery.
Dec	- Set up a development environment, repositories, hosting, database server.
Dec	- Create UI/UX design; Work on the front end; 3D print/make a plant pot.
Jan	- Visualize mock data in a simple dashboard; Setup back-end for storage.
Jan	- Configure hardware components (Raspberry Pi, Sensors, Pumps).
Feb	- Establish a reliable data flow from a single sensor to web and database.
Feb	- Integrate OpenWeather API; Read data from multiple sensors.

Mar	- Develop an autonomous watering system and create an alert system.
Mar	- Conduct full system testing (hardware + software + AI). - Validate output, fix sync and latency issues.
Apr	- Prepare project documentation and presentation materials. - Create a demonstration setup.
Apr	- Finalize all documentation and deliver the final presentation.

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